

Queue Interview Handbook

1. Why was Queue created?

Imagine standing in a ticket counter line. The person who comes first gets served first. This is FIFO (First In, First Out). Queue models this behavior. Common uses include print queues, CPU scheduling, customer support systems, food delivery orders, and message processing.

2. What is a Queue?

A Queue is a collection where elements are added at the rear and removed from the front. Example:

Front -> 10 -> 20 -> 30 -> 40 <- Rear
poll() removes 10 first.

3. Queue Interface

Queue is an interface. Common implementations are LinkedList, PriorityQueue and ArrayDeque.

4. Why is LinkedList commonly used?

LinkedList maintains head and tail references. offer() adds at the tail in $O(1)$. poll() removes from the head in $O(1)$. No element shifting is required.

5. Queue Operations

offer() : Adds an element to the rear.

poll() : Removes and returns the front element. Returns null if empty.

peek() : Returns the front element without removing it. Returns null if empty.

remove() : Same as poll(), but throws an exception if empty.

element() : Same as peek(), but throws an exception if empty.

6. poll() vs remove()

poll()	remove()
Returns null if empty	Throws Exception if empty

7. peek() vs element()

peek()	element()
Returns null if empty	Throws Exception if empty

8. Time Complexity

Operation	Complexity	Reason
offer()	$O(1)$	Tail update
poll()	$O(1)$	Head update
peek()	$O(1)$	Read head
contains()	$O(n)$	Traversal

9. Queue vs Stack

Queue follows FIFO. Stack follows LIFO (Last In, First Out). Queue removes from the front, while Stack removes from the top.

10. Common Interview Questions

Why is Queue commonly implemented using LinkedList? Because insertion at the end and removal from the beginning are both $O(1)$.

Can Queue access the middle element? No, only the front element is directly accessible.

11. Common Mistakes

Thinking Queue allows random access. Using `remove()` without checking for an empty queue.

Assuming Queue always sorts elements (only PriorityQueue does).

12. Remember Forever

Queue = FIFO. `offer()` adds. `poll()` removes. `peek()` views front. LinkedList is a common implementation because head/tail operations are $O(1)$.